
Comprehensive Study Notes

Complete coverage of all course topics

History · Types · Standards · Protocols · Routing · Error Detection
DNS · DHCP · Switching · Topologies · Transmission Media

Compiled for academic use

Table of Contents

Part 1: Foundations

- 1. History of Networks
- 2. Types of Networks
- 3. Network Standards & the OSI Model

Part 2: Physical Layer

- 4. Guided & Unguided Transmission Media
- 5. Types of Cables
- 6. Forms of Communication Lines

Part 3: Protocols & Routing

- 7. Network Protocols
- 8. Network Routing Techniques & Protocols

Part 4: Communication & Data

- 9. Communication Technologies
- 10. Types of Communication
- 11. Components of Data Communication
- 12. Data Flow & Representation
- 13. Switching Techniques

Part 5: Network Management

- 14. DNS — Domain Name System
- 15. DHCP — Dynamic Host Configuration Protocol
- 16. Error Detection & Correction Techniques
- 17. Network Topology
- 18. Criteria for Good Network Performance

Part 1: Foundations

1. History of Networks

The development of computer networks represents humanity's growing need to share information faster and across greater distances. Each era solved a new problem in connectivity.

Era	Milestone	Significance
1960s	ARPANET	US military linked 4 universities — the first packet-switched network
1970s	TCP/IP invented	Vint Cerf & Bob Kahn create the universal language of networks
Early 1980s	Ethernet & LAN rise	Offices begin connecting computers in local networks
1983	TCP/IP becomes standard	ARPANET adopts TCP/IP — the modern internet is born
1991	World Wide Web	Tim Berners-Lee makes the internet accessible to everyone
2000s	Broadband & Wireless	Wi-Fi, mobile networks & cloud computing transform society

Memory Tip

Memory sequence: ARPA → TCP → LAN → Web → Wireless. Each step made networks bigger, faster, and more accessible.

2. Types of Networks

Networks are classified by size and geographic coverage. Think of them as nested layers — each larger type contains the smaller ones.

Type	Full Name	Coverage	Example
PAN	Personal Area Network	~10 metres	Bluetooth devices, smartwatch
LAN	Local Area Network	Building / campus	Office network, school lab
MAN	Metropolitan Area Network	City-wide	City fibre network, cable TV
WAN	Wide Area Network	Countries / global	The Internet

WLAN	Wireless LAN	Building (wireless)	Campus Wi-Fi
VPN	Virtual Private Network	Any distance (secure tunnel)	Remote work access

Memory Tip

Think of it like nesting dolls: PAN fits inside LAN, LAN fits inside MAN, MAN fits inside WAN. The bigger the name, the bigger the network.

3. Network Standards & the OSI Model

Standards are agreed rules that allow devices from different manufacturers to communicate. The key framework is the OSI (Open Systems Interconnection) Model — a 7-layer structure defining how data travels from one device to another.

Standards Organizations:

- **IEEE** — Sets Ethernet (802.3) and Wi-Fi (802.11) standards
- **ISO** — Created the OSI model
- **IETF** — Manages internet protocols like TCP/IP
- **ITU** — International Telecommunications Union; regulates global telecoms

The OSI Model — 7 Layers:

Layer	Name	Function	Protocols/Examples
7	Application	Interface between user and network	HTTP, FTP, SMTP, DNS
6	Presentation	Translates, encrypts & compresses data	SSL/TLS, JPEG, ASCII
5	Session	Opens, manages & closes communication sessions	NetBIOS, RPC
4	Transport	End-to-end delivery, error checking, flow control	TCP, UDP
3	Network	Logical addressing & routing between networks	IP, ICMP, Routers
2	Data Link	Node-to-node transfer, error detection (frames)	Ethernet, MAC, Switches
1	Physical	Raw bit transmission over physical medium	Cables, hubs, voltages

Memory Tip

Mnemonic (bottom to top): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application.

Part 2: Physical Layer

4. Guided & Unguided Transmission Media

The transmission medium is the physical path data travels on. There are two families:

- **Guided (Wired)** — data is confined within a physical cable
- **Unguided (Wireless)** — data propagates as electromagnetic waves through the air

Medium	Type	Bandwidth	Range	Interference	Cost
Twisted Pair	Guided	Medium	Short	High	Low
Coaxial Cable	Guided	Medium	Medium	Medium	Medium
Fiber Optic	Guided	Very High	Very Long	None	High
Radio Waves	Unguided	Low	Very Long	High	Low
Microwave	Unguided	High	Medium (LOS)	Medium	Medium
Infrared	Unguided	Low	Very Short	High	Low

Key Concept

Key insight: Wired = faster, more secure, less interference. Wireless = convenient and mobile, but affected by obstacles and interference. Fiber optic is immune to electromagnetic interference because it uses light, not electricity.

5. Types of Cables

Twisted Pair Cable

The most common cable type. Two copper wires twisted around each other — the twisting cancels electromagnetic interference from adjacent cables. Available in two variants:

- **UTP (Unshielded Twisted Pair)** — common in homes and offices. Cheaper.
- **STP (Shielded Twisted Pair)** — extra metallic shield for industrial use.
- **Categories:** Cat5 (100 Mbps), Cat5e (1 Gbps), Cat6 (10 Gbps), Cat6a, Cat7 — higher categories carry more data.

Coaxial Cable

Has a layered structure: central copper conductor → plastic insulation → metallic shield → outer plastic jacket. This layering blocks interference effectively. Still used in cable TV and some

broadband setups but rare in modern LANs.

Fiber Optic Cable

Transmits light pulses through glass or plastic fibers. Because light travels at extraordinary speed and is unaffected by electromagnetic fields, it is the fastest and most reliable medium. Two types:

- **Single-mode fiber** — very thin core (9 microns), carries one light beam over very long distances. Used in internet backbones and undersea cables.
 - **Multi-mode fiber** — wider core (50–62.5 microns), carries multiple light beams, but over shorter distances. Used within buildings.
-

6. Forms of Communication Lines

Communication lines describe how the physical link between two points is configured and shared:

By connectivity arrangement:

- **Point-to-Point** — a dedicated line connects exactly two devices. Direct, private, and fast — but expensive. Example: a leased line between two company offices.
- **Multi-point (Multi-drop)** — one line is shared among multiple devices. More economical but devices must take turns. Only one device transmits at a time.

By line type:

- **Analog lines (PSTN)** — carry continuously varying signals. Require modems to carry digital data. Traditional telephone network.
- **Digital lines (ISDN, DSL, Fiber)** — carry discrete binary signals natively. Faster and cleaner than analog.
- **Leased lines** — private, permanent connections rented from a telecom provider. Guaranteed dedicated bandwidth, not shared. Used by businesses.
- **Dial-up lines** — connection established on demand by dialing a number. Slow but cheap. Now obsolete.

Part 3: Protocols & Routing

7. Network Protocols

A protocol is a set of agreed rules governing how devices communicate. Without protocols, devices from different manufacturers couldn't understand each other — they are the shared language of networks.

Protocol	Full Name	Purpose	Layer
TCP	Transmission Control Protocol	Reliable, connection-oriented delivery with handshake and acknowledgements	4 (Transport)
UDP	User Datagram Protocol	Fast, connectionless delivery — no handshake, no retransmission	4 (Transport)
IP	Internet Protocol	Logical addressing and routing of packets between networks	3 (Network)
HTTP/HTTPS	Hypertext Transfer Protocol (Secure)	Web page requests and responses; HTTPS adds TLS encryption	7 (Application)
FTP	File Transfer Protocol	Transferring files between computers	7 (Application)
SMTP	Simple Mail Transfer Protocol	Sending email messages	7 (Application)
POP3/IMAP	Post Office / Internet Message Access Protocol	Retrieving email from a server	7 (Application)
DNS	Domain Name System	Translates domain names to IP addresses	7 (Application)
DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP addresses to devices	7 (Application)

Memory Tip

TCP vs UDP: TCP = Trustworthy, Careful, Patient (web, email, files). UDP = Ultra-fast, Don't-wait, Proceed (video, gaming, live calls).

The TCP Three-Way Handshake:

- **Step 1 — SYN:** Sender says 'I want to connect. My sequence number is X.'
- **Step 2 — SYN-ACK:** Receiver replies 'OK. My sequence number is Y. Acknowledged X.'

- **Step 3 — ACK:** Sender confirms 'Acknowledged Y. Connection established.'

After the handshake, data flows and every packet is acknowledged. Lost packets are retransmitted automatically.

8. Network Routing Techniques & Protocols

Routing is the process of finding the best path for a packet to travel from source to destination across a network of routers — like a delivery driver choosing which roads to take.

Static vs Dynamic Routing:

- **Static routing** — an administrator manually programs fixed paths into each router. Simple and predictable, but doesn't adapt if a link fails. Good for small networks.
- **Dynamic routing** — routers automatically discover and update paths using routing protocols. They share network state with each other. When a link fails, they recalculate. This is what the internet uses.

Routing Protocols:

Protocol	Type	Algorithm	Best For	Limitation
RIP	IGP (Interior)	Distance Vector — counts hops	Small networks (max 15 hops)	Slow convergence, limited scale
OSPF	IGP (Interior)	Link State — Dijkstra's algorithm, builds full map	Large enterprise networks	More complex to configure
EIGRP	IGP (Interior)	Hybrid — combines distance vector & link state	Cisco networks	Cisco proprietary
BGP	EGP (Exterior)	Path Vector — policy-based routing between organizations	The internet backbone (ISPs)	Very complex configuration

Memory Tip

RIP = Really Isn't Practical (for large networks). OSPF = Optimal Shortest Path Finder. BGP = Big Global Protocol — it literally runs the internet.

Part 4: Communication & Data

9. Communication Technologies

These are the actual systems and protocols used to send data across networks:

Technology	Standard	Speed / Range	Common Use
Ethernet	IEEE 802.3	10 Mbps – 100 Gbps, wired	Office LANs, data centers
Wi-Fi	IEEE 802.11	Up to several Gbps, wireless	Home and campus networks
Bluetooth	IEEE 802.15	~3 Mbps, 10–100 metres	PAN devices, headsets
3G/4G LTE	3GPP standards	Up to 150 Mbps, city-wide	Mobile internet
5G	3GPP Release 15+	Up to 10 Gbps, city-wide	Next-gen mobile, IoT
DSL	ITU G.992	Up to 100 Mbps, over phone lines	Home broadband
Fiber Broadband	Various	1 Gbps+, very long distances	ISP backbone, premises
Satellite	Various	Up to 100+ Mbps, global coverage	Remote/rural areas

Key Concept

Circuit switching vs Packet switching: Circuit switching reserves a dedicated path for the entire conversation (old telephone system). Packet switching breaks data into packets, each routed independently — this is how the internet works and is far more efficient.

10. Types of Communication

By direction of data flow:

- **Simplex** — one direction only. The receiver can never reply. Example: a TV broadcast, a radio station, a keyboard sending to a computer.
- **Half-duplex** — both directions, but NOT simultaneously. One party talks while the other listens, then they swap. Example: walkie-talkies, push-to-talk radio.
- **Full-duplex** — both directions simultaneously. Both parties can send and receive at the same time. Example: a telephone call, video conferencing.

By number of recipients:

- **Unicast** — one sender to one specific receiver. Example: a private message, a direct phone call.
- **Multicast** — one sender to a specific group of receivers. Example: a group video call, IPTV subscription channels.
- **Broadcast** — one sender to ALL devices on the network. Example: a DHCP Discover message, a radio station.

11. Components of Data Communication

Data communication requires five essential components working together. Remember them with the acronym SMTMP:

Component	Description	Example
Sender (Source)	The device originating the message	Your laptop, smartphone, sensor
Message	The actual data being transmitted (text, audio, video, binary)	An email, a video file, a voice call
Transmission Medium	The physical path the signal travels through	Ethernet cable, fiber optic, radio waves
Receiver (Destination)	The device that receives and processes the message	A server, another phone, a printer
Protocol	The agreed rules both sides follow to understand each other	TCP/IP, HTTP, Bluetooth protocol

Memory Tip

Real-world analogy: Sending a letter. You (sender) write in English (protocol), post it via Royal Mail (medium), and your friend (receiver) reads the message. All five must work together.

12. Data Flow & Data Representation

All digital data ultimately becomes binary — 0s and 1s. Data representation is about how those bits are physically encoded on a medium.

Signal types:

- **Digital signals** — discrete voltage levels. High voltage = 1, low voltage = 0. Used by Ethernet.
- **Analog signals** — continuous waves. Traditional telephone lines used analog. Modems convert digital data to analog for phone lines.

Encoding schemes:

- **NRZ (Non-Return to Zero)** — 1 = high voltage, 0 = low voltage, signal holds constant.
- **Manchester encoding** — each bit has a transition in the middle (1 = high-to-low, 0 = low-to-high). Transitions help receivers sync their clock. Used in Ethernet.
- **NRZI** — signal changes only when a 1 is encountered; holds steady for 0.

Character encoding standards:

- **ASCII** — 7 bits, 128 characters. Covers English letters, numbers, and symbols.
- **Unicode / UTF-8** — covers all human languages, emojis, and special characters. Up to 4 bytes per character.

13. Switching Techniques

Switching is how a network decides to forward data from one device to another. There are three main approaches:

Technique	How it works	Efficiency	Delay	Example
Circuit Switching	A dedicated physical path is reserved for the entire conversation before communication begins	Low — bandwidth wasted when idle	Low after setup	Traditional telephone (PSTN)
Packet Switching	Data is broken into packets; each routed independently through the network and reassembled at destination	High — network shared by many users	Variable	The Internet (TCP/IP)
Message Switching	Entire message stored at each intermediate node, then forwarded. Store-and-forward. Now obsolete.	Medium	Very High	Old telegraph systems

Key Concept

The internet runs on packet switching. Packets from the same message may take completely different routes and arrive out of order — TCP reassembles them in the correct sequence.

Part 5: Network Management

14. DNS — Domain Name System

DNS is the internet's phone book. Humans remember names like google.com; computers use IP addresses like 142.250.185.46. DNS translates one to the other.

The DNS Resolution Process:

- 1. **Check local cache** — browser checks if it has visited this domain recently.
- 2. **Ask the router** — if not cached, the device asks its configured DNS server.
- 3. **Recursive resolver (ISP)** — queries on your behalf if it doesn't know the answer.
- 4. **Root DNS server** — there are 13 root server clusters globally; they point to the TLD server.
- 5. **TLD server** — the .com, .org, .ng servers point to the authoritative server.
- 6. **Authoritative DNS server** — the definitive record for that domain; returns the IP address.
- 7. **Browser connects** — with the IP address in hand, the browser sends an HTTP request.

Key Concept

DNS records are cached at each step to speed up future lookups. The TTL (Time To Live) value controls how long a cached result is kept before re-querying.

15. DHCP — Dynamic Host Configuration Protocol

DHCP automatically assigns IP addresses to devices on a network. Without it, every device would require manual IP configuration. When your phone connects to Wi-Fi, DHCP assigns it: an IP address, a subnet mask, a default gateway, and DNS server addresses.

The DORA Process (4 steps):

Step	Name	Who sends it	What it says
D	Discover	Client (broadcast)	'Is there a DHCP server? I need an IP address!'
O	Offer	DHCP Server	'I offer you IP 192.168.1.5 for 24 hours'
R	Request	Client (broadcast)	'I accept the offer from that server!'
A	Acknowledge	DHCP Server	'Confirmed. IP 192.168.1.5 is yours for 24 hours'

Memory Tip

DORA = Discover, Offer, Request, Acknowledge. Like checking into a hotel: you arrive (Discover), receptionist offers a room (Offer), you accept (Request), they hand you the key (Acknowledge).

16. Error Detection & Correction Techniques

When data travels across a network, bits can flip due to electrical interference, signal degradation, or hardware faults. Error handling techniques detect and sometimes fix these problems.

Error Detection Techniques:

- **Parity Check** — adds one extra bit to each byte so the total number of 1s is always even (even parity) or always odd (odd parity). If a single bit flips, parity is violated. Limitation: two simultaneous bit flips cancel each other and go undetected.
- **Checksum** — the sender sums all data values and sends the result alongside the data. The receiver performs the same calculation and compares. A mismatch = error. Used by TCP/IP.
- **CRC (Cyclic Redundancy Check)** — the most powerful detection technique. Data is divided by a polynomial; the remainder (the CRC value) is appended. Receiver performs the same division — if remainder is zero, no error. Used in Ethernet frames.

Error Correction Techniques:

- **Hamming Code** — inserts multiple parity bits at specific positions (powers of 2: 1, 2, 4, 8...) that create a grid. If any bit flips, the pattern of failed parity checks identifies the exact position of the error, allowing it to be corrected without retransmission. Used in RAM.
- **ARQ (Automatic Repeat reQuest)** — a detection-based system where the receiver requests retransmission of any corrupted packet. Used by TCP.

Technique	Type	Detects	Corrects?	Used In
Parity Check	Detection	Single-bit errors	No	Serial communication
Checksum	Detection	Multiple-bit errors	No	TCP/IP, UDP
CRC	Detection	Burst errors	No	Ethernet, Wi-Fi
Hamming Code	Correction	Single-bit errors	Yes	RAM, deep-space comm
ARQ	Detection + Retransmit	Any detectable error	Via retransmit	TCP

Key Concept

Key distinction: Detection means you know an error happened but cannot fix it. Correction means you can fix it without retransmission. Detection requires the sender to retransmit; correction does not.

17. Network Topology

Topology describes the physical or logical arrangement of devices in a network — the shape of the network. There are two kinds: Physical topology (how devices are actually wired) and Logical topology (how data actually flows).

Topology	Description	Advantages	Disadvantages	Example
Bus	All devices share a single cable (backbone). Data goes to all; only intended recipient accepts.	Cheap, easy to install, uses less cable	Cable failure = entire network down; collisions increase with more devices	Older 10BASE2 Ethernet
Star	All devices connect to a central hub or switch. Devices communicate through the hub only.	Easy to add/remove devices; one cable fails = only that device affected	Hub/switch is single point of failure; needs more cable	Modern office LANs
Ring	Devices connected in a closed loop. Data travels in one direction around the ring.	Equal access; predictable performance; no collisions	One break in the ring stops all communication; slower than star	FDDI, old Token Ring
Mesh	Every device connects directly to every other device. Full mesh = $n(n-1)/2$ connections.	Highly reliable; multiple paths available; no single failure point	Very expensive; complex wiring; difficult to manage	Internet backbone, critical infrastructure
Tree (Hierarchical)	Root node branches into sub-networks. A star of stars — switches connect to a root switch.	Highly scalable; easy to manage; isolates network segments	Root switch failure affects whole network; complex cabling	Large campus networks, ISPs
Hybrid	Combination of two or more topologies. E.g., a tree with ring sub-networks.	Flexible; can be optimized for specific needs	Complex design and maintenance	Most real-world networks

18. Criteria Guaranteeing Good Network Performance

A good network must satisfy five key criteria. Your lecturer may use these as a framework for evaluation questions:

Criterion	Definition	Key Metrics
Performance	How fast and efficiently the network delivers data	Throughput (actual data per second), Latency/Delay (travel time), Jitter (variation in delay)
Reliability	How consistently the network stays up and functional	Uptime % (e.g., 99.9%), MTBF (Mean Time Between Failures), MTTR (Mean Time To Repair), Error rate
Security	Protection of data from unauthorized access, modification, or destruction	Confidentiality, Integrity, Availability (CIA triad), authentication, encryption, firewall policies
Scalability	Ability to grow — adding users, devices, or bandwidth without redesigning	Ease of expansion, performance under load, modular architecture
Quality of Service (QoS)	Ability to prioritize certain traffic types (e.g., voice over file transfer)	Traffic classification, queuing policies, bandwidth reservation, packet prioritization

Memory Tip

Acronym: PRSSC — Performance, Reliability, Security, Scalability, QoS (Service). A network failing on any one of these feels broken to users even if the others are perfect.

Complete Topic Summary

#	Topic	Core Concept to Remember
1	History of Networks	ARPA (1960s) → TCP/IP (1970s) → LAN (1980s) → Web (1991) → Wireless (2000s)
2	Types of Networks	PAN → LAN → MAN → WAN (smallest to largest geographic scope)
3	Network Standards	OSI 7 layers; governed by IEEE, ISO, IETF. Mnemonic: Please Do Not Throw Sausage Pizza Away
4	Transmission Media	Guided (cables) vs Unguided (air). Fiber = fastest, no interference. Radio = long range, interference.
5	Types of Cables	Twisted pair (cheapest), Coaxial (shielded), Fiber optic (light, fastest, most expensive)

6	Forms of Comm. Lines	Point-to-point vs multipoint; analog vs digital; leased vs dial-up
7	Network Protocols	TCP (reliable) vs UDP (fast). IP addresses. HTTP/SMTP/FTP for applications.
8	Routing	Static vs dynamic. RIP (hops), OSPF (map), BGP (internet backbone). IGP vs EGP.
9	Comm. Technologies	Ethernet, Wi-Fi, Bluetooth, 4G/5G, Fiber broadband, Satellite
10	Types of Communication	Simplex/Half-duplex/Full-duplex (direction). Unicast/Multicast/Broadcast (recipients).
11	Components of Data Comm.	Sender, Message, Transmission Medium, Receiver, Protocol — all 5 required.
12	Data Flow & Representation	Digital vs analog signals. Manchester encoding for Ethernet. ASCII and Unicode.
13	Switching Techniques	Circuit (dedicated path), Packet (internet method), Message (store-and-forward, obsolete).
14	DNS	Translates domain names to IP addresses. Process: cache → resolver → root → TLD → authoritative.
15	DHCP	Auto-assigns IPs. DORA: Discover, Offer, Request, Acknowledge.
16	Error Detection	Parity (simple), Checksum (TCP/IP), CRC (Ethernet), Hamming Code (corrects errors, used in RAM).
17	Network Topology	Bus, Star, Ring, Mesh, Tree, Hybrid. Star most common today. Mesh most reliable.
18	Network Performance	Performance, Reliability, Security, Scalability, QoS — all five must be met.

End of ICT 305 Study Notes